High-Uniformity Calculation Method of Four-Coil Configuration in Large-Caliber Magnetic Field Immunity Testing System

Xi Deng, Ya Huang, Chenguang Wan, Li Jiang, Ge Gao, Zhengyi Huang, and Jie Zhang

***Abstract*— Power electronic equipment regulated by International Thermonuclear Experimental Reactor (ITER) must pass the relevant steady-state magnetic field immunity test. The main body of magnetic field immunity test is magnetic field generator coil. Through mathematical derivation in this article, the magnetic field calculation formulas of four-coil configuration under ideal and actual models are obtained. The traditional method of magnetic field performance calculation is compared with the general formula method under the ideal model. A global parameter optimization method based on Lagrange Multiplier by Karush–Kuhn–Tucker (KKT) conditions is proposed to obtain the coil parameters of high-uniformity magnetic field. The magnetic field distribution in the uniform zone is revealed by the finite-element method. The model analysis is proven to be correct and effective by experimental results. The research of this article provides a practical scheme for the coil design with high magnetic field and high-quality uniformity.**

***Index Terms*— Finite-element method (FEM), four coil, global optimization, high uniformity, Lagrange multiplier method by Karush–Kuhn–Tucker (KKT) conditions, large caliber, magnetic field.**

I. Introduction

INTERNATIONAL Thermonuclear Experimental Reactor (ITER) is the largest tokamak experimental reactor under construction in the world. The surrounding area will be covered by a high magnetic field during operation, which will affect the safety and reliability of various electronic and electrical equipment. The steady-state maximum magnetic field of the tokamak building outside the device can reach 200 mT. To ensure safe and stable operation, all power electronic equipment in ITER with a magnetic field greater than 5 mT in the surrounding environment of the hall must pass the relevant static magnetic field immunity test [1].

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Xi Deng and Chenguang Wan are with the Institute of Plasma Physics, Chinese Academy of Sciences, Hefei 230031, China, and also with the Graduate School, University of Science and Technology of China, Science Island Branch, Hefei 230026, China (e-mail: xi.deng@ipp.ac.cn; chenguang.wan@ipp.ac.cn).

Ya Huang, Li Jiang, Ge Gao, Zhengyi Huang, and Jie Zhang are with the Institute of Plasma Physics, Hefei Institutes of Physical Science, Chinese Academy of Sciences, Hefei 230031, China (e-mail: ya.huang@ipp.ac.cn; jiangli@ipp.ac.cn; gg@ipp.ac.cn; hzy@ipp.ac.cn; zhangjie@ipp.ac.cn).

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TABLE I
TEST LEVEL OF THE STEADY MAGNETIC FIELD

Test level	Nominal magnetic field	Min/Max magnetic field
1	7.5 mT	10.5 mT/15 mT
2	15 mT	21 mT/30 mT
3	30 mT	42 mT/60 mT
4	60 mT	84 mT/120 mT
5	120 mT	168 mT/240 mT
6	n	1.4 n / 2 n

According to the distribution of the magnetic field required by ITER report [2], the test is divided into five levels. The nominal magnetic field of each test level and the requirements for testing the space magnetic field are shown in Table I. For each test level, to consider a certain margin, the magnetic field in the whole test space of the equipment under the test (EUT) is 1.4 times higher than the nominal magnetic field. Further, considering that high magnetic field may lead to damage of the EUT, it is required that the maximum magnetic field in the whole test space of the EUT should be twice the nominal magnetic field [3].

The following characterization parameters can be introduced in the uniform zone.

1) Side length of square uniform zone ($2s$).
2) Minimum magnetic field ($B_{\min}$): Minimum value of magnetic field space in the uniform zone.
3) Maximum magnetic field ($B_{\max}$): Maximum value of magnetic field space in the uniform zone.
4) Magnetic field uniformity (η): Ratio of maximum to minimum magnetic field

$$\eta = \frac{B_{\max}}{B_{\min}}. \tag{1}$$

There are many devices for generating uniform magnetic field, including solenoid coil, Helmholtz coil, multicoil group, 3-D orthogonal coil, and so on. Solenoid coil is a common structure, and scientists all over the world have carried out detailed research on the structure, principle, test, and application of the coil [4]–[9]. Helmholtz coil is the most studied coil structure with simple structure. The configuration, principle, and optimization of a Helmholtz coil are described and analyzed in detail in [10]–[18]. For applications that

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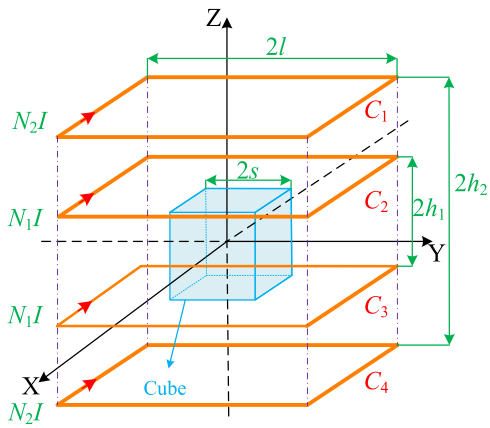

Fig. 1. Four-coil ideal configuration model.

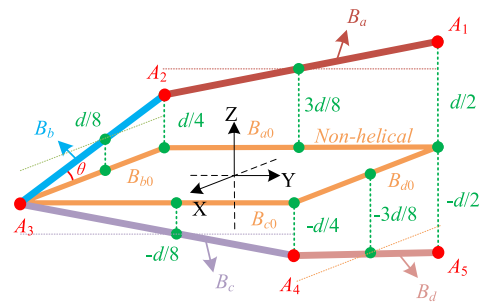

Fig. 2. One turn coil for helical structure.

require a higher magnetic field uniformity, the multicoil configuration can be used to solve the problem. Scientists have studied the magnetic field uniformity of different coil groups and provided reference data in various cases [19]–[31]. The parameters of the multicoil group configuration are optimized and analyzed [31], and the optimal parameters under each structure are obtained. Orthogonal coil configuration also has corresponding references for research and analysis [32], [33].

Because square structure has more advantages than circular structure in the manufacturing and welding process of large equipment [34], [35], and it is recommended by ITER organization [1] and International Electrotechnical Commission (IEC) [36]–[38] in magnetic field immunity test. For the magnetic field immunity testing equipment with test level 4, $B_{\max} = 120$ mT and $\eta \leq 1.2$, and the design, manufacture, and test have been completed [3], [39]. However, the equipment cannot provide help for the requirements of stronger magnetic field and higher uniformity (test level 5, $B_{\text{nom}} = 120$ mT, and $\eta \leq 1.05$). On this basis, this article designs the equipment with higher requirements ($B_{\max} = 275$ mT and $\eta \leq 1.05$).

This article is organized as follows. In Section II, the calculation formulas of uniform regional magnetic field under ideal model and actual model are derived. In Section III, the traditional methods of magnetic field performance calculation and the general formula under the ideal model are compared. Based on Lagrange multiplier method by Karush–Kuhn–Tucker (KKT) conditions, the global parameter optimization is carried out with the actual model to obtain the optimal scheme, and the magnetic field is analyzed by the finite-element method (FEM). In Section IV, experimental verification is provided. The conclusion is given in Section V.

II. Magnetic Field Calculation of Four-Coil Configuration

A. Ideal Model

The four-coil ideal configuration is shown in Fig. 1. The side length of four coils is $2l$, the distance between two inner coils is $2h_1$, the distance between two outer coils is $2h_2$, the ampere turns of two inner coils are $N_1 I$, and the ampere turns of two outer coils are $N_2 I$.

According to Biot-Savart's law and the magnetic field superposition principle, the magnetic field $B_I\ (x,\ y,\ z)$ produced by the square four-coil ideal configuration at the point $P\ (x,\ y,\ z)$ in the uniform zone can be expressed as

$$\begin{cases} B_I(x) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \sum_{t=1}^{2} \left(N_t \cdot \sum_{i=0}^{1} \sum_{j=0}^{1} \sum_{k=0}^{1} \dfrac{(-1)^{i+1} Y_1 \cdot Z_t}{X_1^2 + Z_t^2} \cdot D_t \right) \\ B_I(y) \\ = \dfrac{\mu_0 I}{4\pi} \\ \cdot \sum_{t=1}^{2} \left(N_t \cdot \sum_{i=0}^{1} \sum_{j=0}^{1} \sum_{k=0}^{1} \dfrac{(-1)^{j+1} X_1 \cdot Z_t}{Y_1^2 + Z_t^2} \cdot D_t \right) \\ B_I(z) \\ = \dfrac{\mu_0 I}{4\pi} \\ \cdot \sum_{t=1}^{2} \left(N_t \cdot \sum_{i=0}^{1} \sum_{j=0}^{1} \sum_{k=0}^{1} \left[\dfrac{X_1 \cdot Y_1}{X_1^2 + Z_t^2} + \dfrac{X_1 \cdot Y_1}{Y_1^2 + Z_t^2} \right] \cdot D_t \right) \\ B_I(x, y, z) \\ = \left(B_I^2(x) + B_I^2(y) + B_I^2(z) \right)^{1/2} \end{cases} \tag{2}$$

where $|x| \leq s$, $|y| \leq s$, $|z| \leq s$, $X_1 = (-1)^i x + l$, $Y_1 = (-1)^j y + l$, $Z_t = (-1)^k z + h_t$, and $D_t = (X_1^2 + Y_1^2 + Z_t^2)^{-1/2}$.

The turn ratio of the side coil to the intermediate coil is defined as β and can be described as

$$\beta = N_2 / N_1. \tag{3}$$

The magnetic field value and uniformity in the uniform zone are the functions of parameters l, h_1, h_2, β, s, and $N_1 I$

$$\begin{cases} B_{\max} = f_1(x, y, z | l, h_1, h_2, \beta, s) \cdot N_1 I \\ B_{\min} = f_2(x, y, z | l, h_1, h_2, \beta, s) \cdot N_1 I \\ \eta = g(x, y, z | l, h_1, h_2, \beta, s). \end{cases} \tag{4}$$

B. Actual Model

Compared with the ideal model, the actual model has the problems of spiral structure and conductor section.

1) Helical Structure: The helical structure is analyzed from one turn coil. The one turn coil is shown in Fig. 2, where θ is the offset angle of one side length, $2l$ is the side length of the primary coil, and d is the distance between each turn of the coil, and its relationship is

$$\theta = \arctan \frac{d}{8l}. \tag{5}$$

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The magnetic field generated by nonhelical one turn coil at the point P (x, y, z) [40] in the uniform zone is

$$\begin{cases} B_{a0}(x) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{-z}{(l+x)^2+z^2} \left(\dfrac{l-y}{\left[(l+x)^2+(l-y)^2+z^2\right]^{1/2}} + \dfrac{l+y}{\left[(l+x)^2+(l+y)^2+z^2\right]^{1/2}} \right) \\ B_{a0}(y)=0 \\ B_{a0}(z) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{l+x}{(l+x)^2+z^2} \left(\dfrac{l-y}{\left[(l+x)^2+(l-y)^2+z^2\right]^{1/2}} + \dfrac{l+y}{\left[(l+x)^2+(l+y)^2+z^2\right]^{1/2}} \right) \end{cases} \tag{6}$$

$$\begin{cases} B_{b0}(x)=0 \\ B_{b0}(y) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{-z}{(l+y)^2+z^2} \left(\dfrac{l+x}{\left[(l+x)^2+(l+y)^2+z^2\right]^{1/2}} + \dfrac{l-x}{\left[(l-x)^2+(l+y)^2+z^2\right]^{1/2}} \right) \\ B_{b0}(z) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{l+y}{(l+y)^2+z^2} \left(\dfrac{l+x}{\left[(l+x)^2+(l+y)^2+z^2\right]^{1/2}} + \dfrac{l-x}{\left[(l-x)^2+(l+y)^2+z^2\right]^{1/2}} \right) \end{cases} \tag{7}$$

$$\begin{cases} B_{c0}(x) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{z}{(l-x)^2+z^2} \left(\dfrac{l-y}{\left[(l-x)^2+(l-y)^2+z^2\right]^{1/2}} + \dfrac{l+y}{\left[(l-x)^2+(l+y)^2+z^2\right]^{1/2}} \right) \\ B_{c0}(y)=0 \\ B_{c0}(z) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{l-x}{(l-x)^2+z^2} \left(\dfrac{l-y}{\left[(l-x)^2+(l-y)^2+z^2\right]^{1/2}} + \dfrac{l+y}{\left[(l-x)^2+(l+y)^2+z^2\right]^{1/2}} \right) \end{cases} \tag{8}$$

$$\begin{cases} B_{d0}(x)=0 \\ B_{d0}(y) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{z}{(l-y)^2+z^2} \left(\dfrac{l-x}{\left[(l-x)^2+(l-y)^2+z^2\right]^{1/2}} + \dfrac{l+x}{\left[(l+x)^2+(l-y)^2+z^2\right]^{1/2}} \right) \\ B_{d0}(z) \\ = \dfrac{\mu_0 I}{4\pi} \cdot \dfrac{l-y}{(l-y)^2+z^2} \left(\dfrac{l-x}{\left[(l-x)^2+(l-y)^2+z^2\right]^{1/2}} + \dfrac{l+x}{\left[(l+x)^2+(l-y)^2+z^2\right]^{1/2}} \right). \end{cases} \tag{9}$$

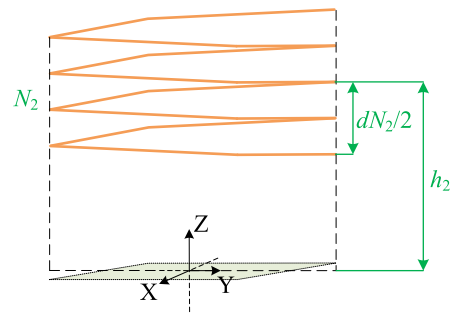

Fig. 3. Location diagram of C_1 coil.

On the basis of the original conductor (B_{a0}), translate the distance of $3d/8$ along the positive direction of the Z-axis and then rotate the angle of θ along the YZ-direction around the X-axis to obtain the conductor B_a. The conductors B_b, B_c, and B_d are obtained by the same conversion method.

The magnetic field generated by the nonhelical conductor is obtained by the coordinate transformation of translation and rotation and then decomposed into the X-, Y-, and Z-axis components of the original coordinate system

$$\begin{cases} [B_a(x)B_a(y)B_a(z)]^{\mathrm{T}} = \boldsymbol{A}[B_{a0}(x_a)B_{a0}(y_a)B_{a0}(z_a)]^{\mathrm{T}} \\ [B_b(x)B_b(y)B_b(z)]^{\mathrm{T}} = \boldsymbol{B}[B_{b0}(x_b)B_{b0}(y_b)B_{b0}(z_b)]^{\mathrm{T}} \\ [B_c(x)B_c(y)B_c(z)]^{\mathrm{T}} = \boldsymbol{C}[B_{c0}(x_c)B_{c0}(y_c)B_{c0}(z_c)]^{\mathrm{T}} \\ [B_d(x)B_d(y)B_d(z)]^{\mathrm{T}} = \boldsymbol{D}[B_{d0}(x_d)B_{d0}(y_d)B_{d0}(z_d)]^{\mathrm{T}} \end{cases} \tag{10}$$

$$\begin{cases} [x_a y_a z_a]^{\mathrm{T}} = \boldsymbol{C}[xyz]^{\mathrm{T}} + [003d/8]^{\mathrm{T}} \\ [x_b y_b z_b]^{\mathrm{T}} = \boldsymbol{D}[xyz]^{\mathrm{T}} + [00d/8]^{\mathrm{T}} \\ [x_c y_c z_c]^{\mathrm{T}} = \boldsymbol{A}[xyz]^{\mathrm{T}} + [00{-}d/8]^{\mathrm{T}} \\ [x_d y_d z_d]^{\mathrm{T}} = \boldsymbol{B}[xyz]^{\mathrm{T}} + [00{-}3d/8]^{\mathrm{T}} \end{cases} \tag{11}$$

where

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 0 & \cos\theta & \sin\theta \\ 1 & 0 & 0 \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 & \cos\theta & -\sin\theta \\ 1 & 0 & 0 \\ 0 & \sin\theta & \cos\theta \end{bmatrix}.$$

According to the magnetic field superposition principle, the magnetic field generated by the point P (x, y, z) in the uniform zone of the intermediate one turn coil is

$$\begin{cases} B_0(x) = B_a(x) + B_b(x) + B_c(x) + B_d(x) \\ B_0(y) = B_a(y) + B_b(y) + B_c(y) + B_d(y) \\ B_0(z) = B_a(z) + B_b(z) + B_c(z) + B_d(z). \end{cases} \tag{12}$$

For the C_1 coil in Fig. 1, the location diagram is shown in Fig. 3, and there are N_2 turns in total. The magnetic field generated by the coil with other turns is equivalent to the translation on the Z-axis. r_1 represents the Nth turn of the C_1 coil. The transformation formula is

$$\left[x_{r_1} y_{r_1} z_{r_1}\right]^{\mathrm{T}} = [xyz]^{\mathrm{T}} + \left[00\left(h_2 - \frac{d(N_2-1)}{2} + r_1 \cdot d\right)\right]^{\mathrm{T}}. \tag{13}$$

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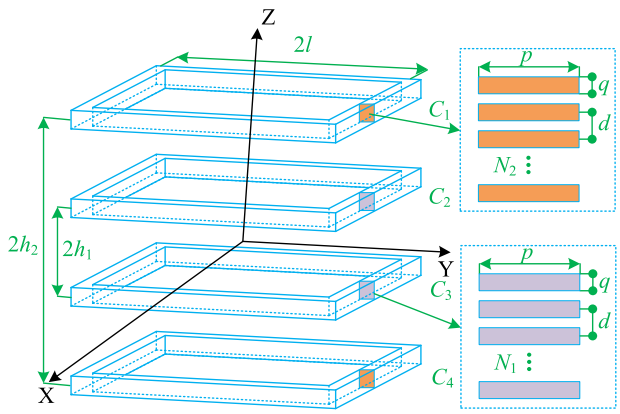

Fig. 4. Cross-sectional models.

The magnetic field generated by N_2 turns in C_1 is

$$\begin{cases} B_{C1}(x) = \sum_{r_1=0}^{N_2} B_0\left(x_{r_1}\right) \\ B_{C1}(y) = \sum_{r_1=0}^{N_2} B_0\left(y_{r_1}\right) \\ B_{C1}(z) = \sum_{r_1=0}^{N_2} B_0\left(z_{r_1}\right). \end{cases} \tag{14}$$

Similarly, the magnetic field generated by other coil groups of four-coil configuration can be obtained, and the total magnetic field B_H (x, y, z) is obtained by superposition

$$\begin{cases} B_H(x) = B_{C1}(x) + B_{C2}(x) + B_{C3}(x) + B_{C4}(x) \\ B_H(y) = B_{C1}(y) + B_{C2}(y) + B_{C3}(y) + B_{C4}(y) \\ B_H(z) = B_{C1}(z) + B_{C2}(z) + B_{C3}(z) + B_{C4}(z) \\ B_H(x, y, z) = \left(B_H^2(x) + B_H^2(y) + B_H^2(z)\right)^{1/2}. \end{cases} \tag{15}$$

2) Cross Section: The shape of the coil conductor cross section has a certain impact on the magnetic field performance [17], [18]. The cross-sectional model is shown in Fig. 4.

The cross-sectional length L of the four-coil groups is $[L_1, L_2]$, $L_1 = l - p/2$ and $L_2 = l + p/2$.

The height H_{1r} of C_1 coil is $[H_1 + r_1 d,\ H_1 + q + r_1 d]$, $H_1 = h_2 - (d \cdot (N_2 - 1) - q)/2$ and $r_1 = \{0, 1, \ldots, N_2\}$.

The height H_{2r} of C_2 coil is $[H_2 + r_2 d,\ H_2 + q + r_2 d]$, $H_2 = h_1 - (d \cdot (N_1 - 1) - q)/2$ and $r_2 = \{0, 1, \ldots, N_1\}$.

The height H_{3r} of C_3 coil is $[H_3 + r_3 d,\ H_3 + q + r_3 d]$, $H_3 = -h_1 - (d \cdot (N_1 - 1) - q)/2$ and $r_3 = \{0, 1, \ldots, N_1\}$.

The height H_{4r} of C_4 coil is $[H_4 + r_4 d,\ H_4 + q + r_4 d]$, $H_4 = -h_2 - (d \cdot (N_2 - 1) - q)/2$ and $r_4 = \{0, 1, \ldots, N_2\}$.

The magnetic field distribution in the uniform zone can be expressed as (16), as shown at the bottom of the next page, where $u_1 = x$, $u_2 = y$, and $u_3 = z$.

Comprehensive consideration shows that the magnetic field performance is related to parameters, such as l, h_1, h_2, d, p, q, s, $N_1 I$, and $N_2 I$, which can be expressed as

$$\begin{cases} B_{\max} = f_1(x, y, z|l, h_1, h_2, d, p, q, s, N_1, N_2) \cdot I \\ B_{\min} = f_2(x, y, z|l, h_1, h_2, d, p, q, s, N_1, N_2) \cdot I \\ \eta = g(x, y, z|l, h_1, h_2, d, p, q, s, N_1, N_2). \end{cases} \tag{17}$$

III. Research of the High-Uniformity Magnetic Coil Design Method

A. Traditional Method

The traditional method is to use Taylor expansion to calculate the coil structure parameters when the higher order of the axis is zero [17], [35], [41]. The axial magnetic field is equal, and the calculation is relatively simple, but it is not optimal for the case that the uniform zone is a cube

$$B(z) = B_z(0) + \sum_{k=1}^{\infty} B_z^{2k}(0) \frac{z^{2k}}{(2k)!}. \tag{18}$$

The parameter relationship of Merritt four-coil configuration is obtained as [35]

$$\begin{cases} h_1 = 0.256212 \cdot l \\ h_2 = 1.010984 \cdot l \\ \beta = 2.361197 \\ N_2 I = 1.1142 \cdot l \cdot B(0) \times 10^6. \end{cases} \tag{19}$$

B. General Formula Method

For the ideal model, the optimized general formulas of four-coil structure and magnetic field performance parameters can be summarized as [30]

$$\begin{cases} l/s = 1.267 + 137.6 \cdot e^{-5.874\eta} \\ h_1/s = 0.1616 \cdot l/s - 0.1594 \\ h_2/s = 1.1942 \cdot l/s - 0.3880 \\ \beta = 0.9779 + 14.19 \cdot e^{-2.375\eta} \\ N_1 I + N_2 I = \left(1.964 + 458.9 \cdot e^{-6.682\eta}\right) s B_{\min} \times 10^6. \end{cases} \tag{20}$$

The calculation of general formula is convenient and fast, but there are still several problems.

1) The calculation range of magnetic field uniformity is $1.1 \le \eta \le 1.4$, which cannot meet the requirements of high-uniformity ($\eta \le 1.05$) magnetic field performance.
2) The influence of helical structure and conductor cross section is not considered in the calculation.
3) Only the axial magnetic field is considered in the calculation, and the magnetic field of the actual model exists on the three axes (X, Y, Z), without considering the proportion of the magnetic field on each axis.

C. Basic Principle of Lagrange Multiplier Method by KKT Conditions

This article mainly considers the optimization of multiparameters. The generalized Lagrange multiplier method by KKT conditions is a good choice for it.

The Lagrange multiplier method is widely used for solving constrained optimization problems. For optimization problems with equality constraints, the Lagrange multiplier method can be used to find the optimal value; if there are inequality constraints, the KKT conditions can be used to generalize the

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Lagrange multiplier method to find the optimal value. Generally, the optimization problem with inequality constraints is solved as follows:

$$\begin{cases} \min f(x) \\ \text{s.t. } h_i(x) = 0, \quad (i = 1, 2, \ldots, m) \\ \quad g_j(x) \le 0, \quad (j = 1, 2, \ldots, n) \end{cases} \tag{21}$$

where $f(x)$ is the original objective function, $h_i(x)$ and $g_j(x)$ are the equality and inequality constraint functions, respectively, and $x = [x_1, x_2, x_3, \ldots,]^{\mathrm{T}}$ is the independent variable.

The Lagrange function is defined as

$$L(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i h_i(x) + \sum_{j=1}^{n} \mu_j g_j(x) \tag{22}$$

where $\lambda = [\lambda_1, \lambda_2, \lambda_{3,\ldots,} \lambda_i]^{\mathrm{T}}$ and $\mu = [\mu_1, \mu_2, \mu_{3,\ldots,} \mu_j]^{\mathrm{T}}$ is the Lagrange multiplier of $h_i(x)$ and $g_j(x)$, respectively.

For the Lagrange multiplier method under simultaneous constraints of inequality and equality, the condition of the optimal solution can be expressed by the following KKT conditions:

$$\begin{cases} \nabla_x L(x^*, \lambda, \mu) = 0 \\ h_i(x^*) = 0 \\ g_j(x^*) \le 0 \\ \mu_j > 0 \\ \mu_j \cdot g_j(x^*) = 0 \\ \nabla_\lambda L(x^*, \lambda, \mu) = 0 \\ \nabla_\mu L(x^*, \lambda, \mu) = 0 \end{cases} \tag{23}$$

where x^* is the extreme point of (21).

D. Global Optimization Method—Lagrange Multiplier Method by KKT Conditions

The optimization model is constructed as follows.

1) Part I: The optimization objectives are described.

The optimal state is the minimum weight and power loss. The total weight G and power loss P can be expressed as

$$\begin{cases} G = \rho V = 16\dfrac{\rho}{J} \cdot (N_1 + N_2) \cdot I \cdot l \\ P = \dfrac{J^2 V}{\sigma} = 16\dfrac{J}{\sigma} \cdot (N_1 + N_2) \cdot I \cdot l \end{cases} \tag{24}$$

where ρ is the conductor mass density, J is the conductor current density, σ is the conductor conductivity, and V is the total volume.

When the parameter $A = (N_1 + N_2) \cdot I \cdot l$ is the smallest, the weight and power loss are the smallest, which can be used as the optimization goal.

2) Part II: The equality and inequality constraints are considered.

According to the design requirement, the magnetic field $B_{\max} = 275$ mT. The magnetic field uniformity η *is* 1.05. The size of the test zone is $1 \times 1 \times 1$ m^3. A space of 0.325 m is reserved at each side for the installation of the EUT handling platform; thus, the inner space $[a = 2 \times (l - p/2)]$ of the test coil should be 1.65 m at least. The separation distance of the inner two coils ($d_2 = 2 \times H_2$) is 0.2 m. The separation distance of the inner coil and the outer coil $[d_1 = H_1 - (H_2 + q + r_2 d)]$ is in the range of 120–200 mm. The spacing d between each turn of the coil should be less than 10 mm. The current density cannot exceed 2.5 A/mm^2. Limited by the output power of the power supply, the current per turn of the coil shall not exceed 12.5 kA.

Thus, the optimization model is shown as follows:

$$\min A = (N_1 + N_2) \cdot I \cdot l$$
$$\text{s.t.} \begin{cases} s = 1 \text{ m} \\ d_2 = 200 \text{ mm} \\ \eta \le 1.05 \\ I \le 12.5 \text{ kA} \\ d \le 10 \text{ mm} \\ 120 \text{ mm} \le d_1 \le 200 \text{ mm} \\ J \le 2.5 \text{ A/mm}^2 \\ a = 2 \times (l - p/2) \ge 1650 \text{ mm} \\ B_{\max} = 275 \text{ mT} \end{cases} \tag{25}$$

where

$$\begin{cases} B_{\max} = f_1(x, y, z | l, d_1, d_2, p, q, d, s, N_1, N_2) \cdot I \\ B_{\min} = f_2(x, y, z | l, d_1, d_2, p, q, d, s, N_1, N_2) \cdot I \\ \eta = g(x, y, z | l, d_1, d_2, p, q, d, s, N_1, N_2). \end{cases} \tag{26}$$

Based on (5)–(17) and (21)–(26), one certain group of optimized coil parameters with the minimum value of $A = (N_1 + N_2) \cdot I \cdot l$ can be obtained for a specific homogeneous

$$B_{pq} = \frac{1}{pq} \left(\sum_{t=1}^{3} \left(\begin{array}{l} \displaystyle\sum_{r_1=0}^{N_2} \left(\int_{L_1}^{L_2} \int_{H_1+r_1 d}^{H_1+r_1 d+q} B_{C1}(u_t | L, H_{1r}) \mathrm{d}L \mathrm{d}H_{1r} \right) \\ \displaystyle + \sum_{r_2=0}^{N_1} \left(\int_{L_1}^{L_2} \int_{H_2+r_2 d}^{H_2+r_2 d+q} B_{C2}(u_t | L, H_{2r}) \mathrm{d}L \mathrm{d}H_{2r} \right) \\ \displaystyle + \sum_{r_3=0}^{N_1} \left(\int_{L_1}^{L_2} \int_{H_3+r_3 d}^{H_3+r_3 d+q} B_{C3}(u_t | L, H_{3r}) \mathrm{d}L \mathrm{d}H_{3r} \right) \\ \displaystyle + \sum_{r_4=0}^{N_2} \left(\int_{L_1}^{L_2} \int_{H_4+r_4 d}^{H_4+r_4 d+q} B_{C4}(u_t | L, H_{4r}) \mathrm{d}L \mathrm{d}H_{4r} \right) \end{array} \right)^2 \right)^{\frac{1}{2}} \tag{16}$$

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TABLE II
PARAMETERS OF THE FOUR-COIL SYSTEM

Parameter	η	B_{min}	B_{max}	I	N_1	N_2	s
Value	1.0495	262.03	275	12.23	20	12	1000
Unit	--	mT	mT	kA	--	--	mm
Parameter	$2l$	d_1	d_2	d	p	q	
Value	2190	140	200	8	270	27	
Unit	mm	mm	mm	mm	mm	mm	

field. Since the model is a symmetrical structure, only 1/8 of the structure is calculated.

After applied the KKT, we are given a normal optimization problem. A descent method was generally used for normal optimization problem consisting of the following steps.

1) Choose an initial feasible solution $x = [x_1, x_2, x_3, \ldots,]^T$. Here, we initialize the parameters using the result of Taylor expansion in (18) and (19) and previous engineering experience.
2) Identify a feasible "target" solution x^D in a "downhill direction." with momentum term.
3) Choose a step size $\alpha = 0.01$, and set $x = \alpha x^D + (1 - \alpha)\, x$.
4) Test for termination, and return to step (2) if we need to improve further.

After the above steps, we can get the optimal value. The calculation result is as follows. $I = 12.23$ kA, $N_1 = 19.82$, $N_2 = 12.11$, $l = 1093$ mm, $d_1 = 140.80$ mm, $d_2 = 200.55$ mm, $d = 8$ mm, $p = 268.8$ mm, and $q = 27.2$ mm. Since the number of turns is an integer, N_1 and N_2 are selected as 20 and 12, and considering the accuracy of engineering fabrication, the values of other parameters are shown in Table II.

E. Calculation of FEM

In this article, finite-element software is used for the simulation calculation. The simulation model of the four-coil system is under an actual state, where the conductor cross-sectional size and the helical structure are considered. $\eta = 1.05$ and $B_{min} = 275$ mT are taken as an example. According to the above design parameters, the simulation model and results of the actual model are obtained, as shown in Fig. 5.

The data from the FEM simulation are compared with the analytical method calculation, as shown in Table III. The calculation error is less than 0.428%. The final simulation result is $\eta = 1.054$. Although the value is slightly larger than 1.05 of the design requirements, it is still within 1.1 of the standard requirements. The design meets the requirements.

The magnetic field distribution of each surface in the optimized test area is intercepted. The intercepted surfaces are shown in Fig. 6, which are $z = 0$, $z = 0.5s$, $z = s$, $y = 0$, $y = 0.5s$, and $y = s$, respectively. The x-axis and y-axis are symmetrical in terms of magnetic field, and only one of them need to be considered.

Through FEM simulation, the magnetic field distribution on the six surfaces is calculated, as shown in Fig. 7. It can be seen

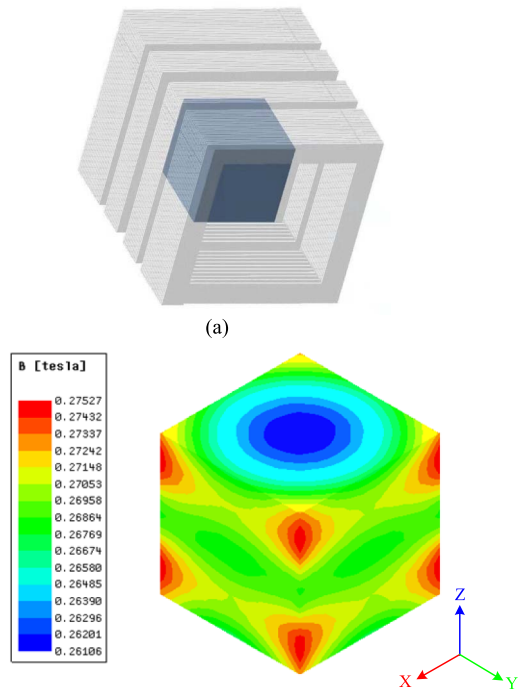

Fig. 5. Simulation model and results of the actual model. (a) Simulation model. (b) Distribution of the magnetic-field after optimization.

TABLE III
ERROR COMPARISON BETWEEN THE ANALYTIC METHOD AND THE FEM

Structure	B_{max}/mT	B_{min}/mT	η
Analytic Method	275.00	262.03	1.0495
FEM	275.27	261.06	1.054
Error	-0.098%	0.370%	-0.428%

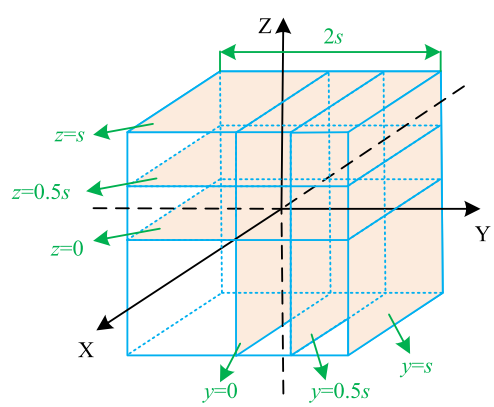

Fig. 6. Cut surface distribution of test area.

that the magnetic field distribution inside the test area, in which the closer to the $z = 0$ plane in the axial direction (z-axis direction), the more uniform the magnetic field distribution. Compared with the z plane, the magnetic field in the y plane changes more violently, and the magnetic field fluctuation is the largest on the $y = s$ plane.

Similarly, calculate the magnetic field distribution on six line segments ($x = 0$, $y = 0$), ($x = 0$, $y = 0.5s$), ($x = 0.5s$,

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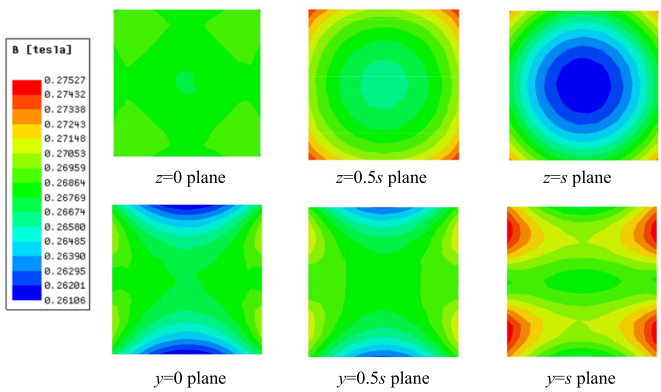

Fig. 7. Magnetic field distribution on different surfaces of the test area after optimization.

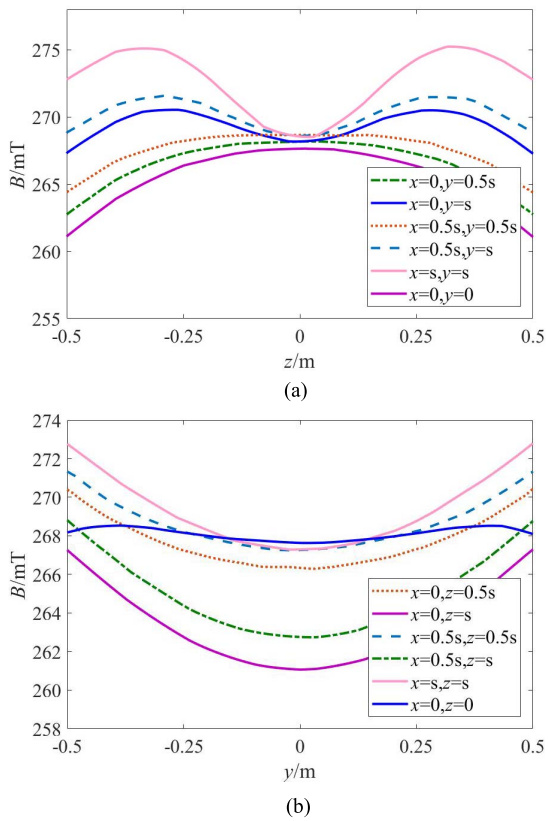

Fig. 8. Magnetic field distribution of different line segments in the test area, after optimization. (a) z-axis direction. (b) y-axis direction.

$y = 0.5s$), ($x = 0$, $y = s$), ($x = 0.5s$, $y = s$), and ($x = s$, $y = s$) in the z-axis direction and six line segments ($x = 0$, $z = 0$), ($x = 0$, $z = 0.5s$), ($x = 0.5s$, $z = 0.5s$), ($x = 0$, $z = s$), ($x = 0.5s$, $z = s$), and ($x = s$, $z = s$) in the y-axis direction, as shown in Fig. 8. In the z-axis direction, the closer to the outer side, the greater the magnetic field fluctuation, and each line is almost close to a point ($z = 0$), indicating that the magnetic field uniformity of the z plane where this point is located is high. In the y-axis direction, the fluctuation of each line itself is not large, but the magnetic field values between lines are quite different.

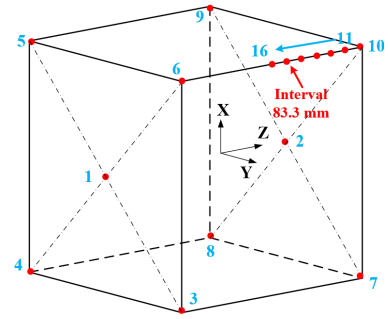

Fig. 9. Layout diagram of magnetic field measuring probe in uniform zone.

IV. Verification

A. Experimental Test

To verify the field inhomogeneity of the magnetic field generated by the test coil, the first step is to position the test zone. A frame structure should be manufactured first to install the probes. Eight probes (3–10 in Fig. 9) located at the corners are used to position the test zone. If a coordinate system is defined at the center of the test zone as shown in Fig. 10, the coordinates of these two points will be (0, 0, −0.5) and (0, 0, 0.5). As the maximum magnetic field occurs on the side edges of the test zone and its position is (0.5, 0.5, 0.32), seven probes (10–16) with a separation distance of 83.3 mm are used to measure the field distribution on a half of the edge. To measure the correct maximum value, the position of probe 12 can be adjusted to (0.5, 0.5, 0.32). The maximum field value will be obtained, and these values will be compared with the calculated ones to verify the theoretical results. It should be pointed out that all the probes are used to measure the axial magnetic field (Z-direction in Fig. 9).

Based on the analysis presented above, 16 Hall probes are employed to calibrate the test field. The probes can be divided into three groups, which are given below.

1) Two probes (1 and 2) to measure the minimum magnetic field.
2) Eight probes (3–10) to locate the test zone.
3) Six probes (11–16) along with 10 to measure the field distribution on the edge and the maximum magnetic field.

The testing model of the four-coil system and the testing system experimental platform is shown in Fig. 10. The system is insulated and fixed by the epoxy board, the water-cooled runner at the coil corner is connected by a water pipe, and the bottom is supported by the bottom foot. The deionized water is flown into each of the turn coils and leads at 1 m/s. The installation position accuracy of the Hall probe shall be within 1.5 mm from the designed locations.

When 12.23-kA current is applied, the magnetic-flux density measured by 16 Hall probes is compared with the simulation values, as shown in Tables IV and V. Table IV shows that the errors are within 2% (standard error is less than 2%) and meet the requirements. The error comparison of $B_{\max}$, $B_{\min}$, and η between the measurement and the simulation is shown in

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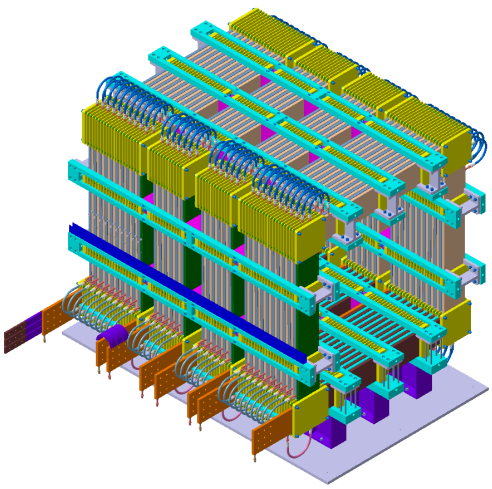
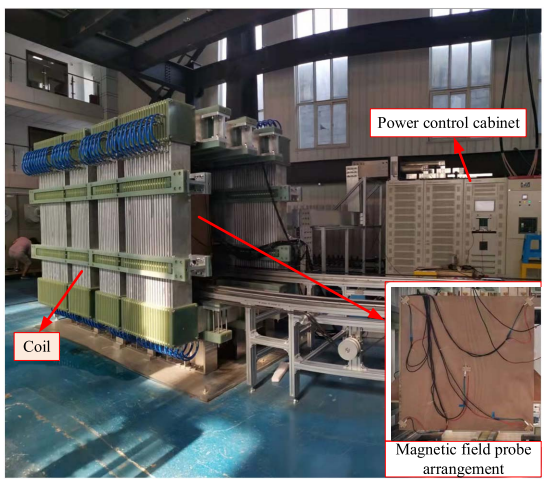


(a)



(b)

Fig. 10. Testing experimental. (a) Testing model of the four-coil system. (b) Test system experimental platform.

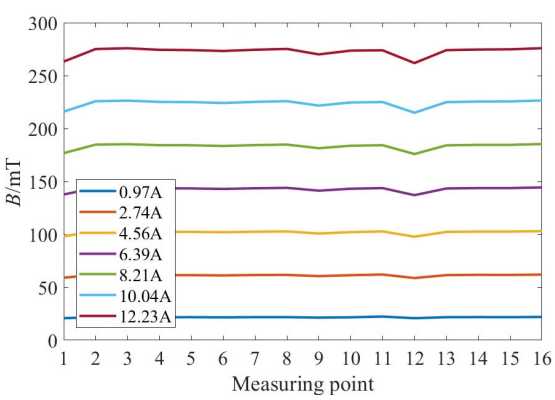


Fig. 11. Measured magnetic field distribution at different current levels.

Table V. The error of the magnetic field uniformity is 0.333%, which meets the requirements.

The measured magnetic field distribution at different current levels is shown in Fig. 11. It can be seen from the picture that points 1 and 2 are the lowest points of the magnetic field at different current levels. Points 11–16 show relatively large magnetic field intensity. Fig. 12 shows that the current

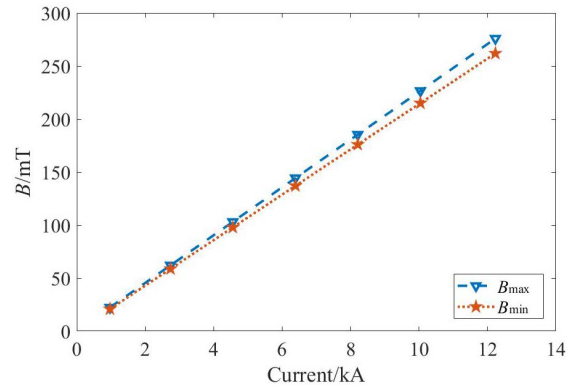


Fig. 12. B_{max} and B_{min} on different current levels.

TABLE IV
ERROR COMPARISON BETWEEN THE MEASUREMENT AND THE FEM

Point	B/mT (FEM)	B/mT (Measurement)	Error
1	261.07	263.26	-0.84%
2	272.81	275.11	-0.84%
3	272.79	275.87	-1.13%
4	272.76	274.38	-0.59%
5	272.76	274.10	-0.49%
6	268.42	273.28	-1.81%
7	269.53	274.42	-1.81%
8	272.00	275.24	-1.19%
9	274.24	270.05	1.53%
10	275.11	273.62	0.54%
11	274.47	273.91	0.20%
12	261.02	261.86	-0.32%
13	272.82	274.07	-0.46%
14	272.80	274.62	-0.67%
15	272.79	274.86	-0.76%
16	272.79	275.90	-1.14%

TABLE V
ERROR COMPARISON THE MEASUREMENT AND THE FEM AT 12.23 KA

	B_{max}/mT	B_{min}/mT	η
Analytic Method	275.00	262.03	1.0495
Measurement	275.89	261.86	1.0530
Error	-0.323%	0.065%	0.333%

is almost linear with the maximum and minimum magnetic field value. Both Figs. 11 and 12 show that the there is little difference between the maximum and minimum magnetic field at the test and the magnetic field uniformity is high.

V. CONCLUSION

Based on ITER's high-uniformity magnetic field immunity testing system, the magnetic field calculation of four-coil configuration is studied. First, the magnetic field calculation formulas of four-coil configuration under ideal and actual

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models are derived. The two main influencing factors of actual coil relative to ideal coil are considered, including helical structure and conductor cross section. Then, a global parameter optimization method based on Lagrange multiplier by KKT conditions is proposed to obtain the coil parameters of high-uniformity magnetic field. Finally, the correctness of the calculation is verified by the experiments of the existing equipment, and the coil structures with high uniformity are compared by using the FEM. Theoretical analysis, finite-element simulation, and experimental data are in good agreement. The proposed method provides a theoretical basis for the subsequent actual manufacturing.

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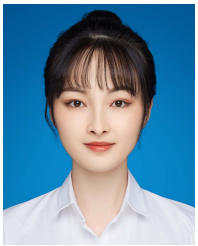

Xi Deng was born in Anhui, China, in 1995. She received the B.S. degree in electrical engineering and automation from Anhui Agricultural University, Hefei, China, in 2017. She is currently pursuing the Ph.D. degree with the University of Science and Technology of China, Hefei. She majored in electronics and magnetic field analysis.

Her current research interests include fault diagnosis of high-power converters and electromagnetic compatibility (EMC) analysis.

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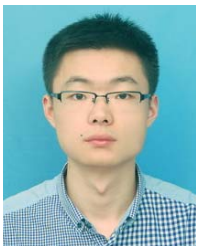
Ya Huang was born in Anhui, China, in 1990. He received the B.S. and M.S. degrees in electrical engineering from the Hefei University of Technology, Hefei, China, in 2011 and 2014, respectively, and the Ph.D. degree in electronics, magnetic field analysis, and thermal analysis from the University of Science and Technology of China, Hefei, in 2021.

His current research interests include electromagnetic compatibility (EMC) analysis and electric machine design.



Chenguang Wan was born in Anhui, China, in 1995. He received the B.S. degree in mechanical design from the Hefei University of Technology, Hefei, China, in 2017. He is currently pursuing the Ph.D. degree with the University of Science and Technology of China, Hefei.

His current research interests include tokamak modeling algorithm design and the intersectional field of machine learning and modeling control.

Li Jiang was born in Anhui, China, in 1981. He received the B.S. degree in electrical motors and automation from Anhui Polytechnic University, Wuhu, China, in 2005, and the Ph.D. degree in nuclear engineering from the Chinese Academy of Sciences, Hefei, China, in 2011.

He is currently an Associate Professor with the International Thermonuclear Experimental Reactor Project, Institute of Plasma Physics, Chinese Academy of Sciences. He majored in electronics, magnetic field analysis, and integration design of high-power supply systems. His current research interests include power supply systems of fusion devices.

Ge Gao was born in Anhui, China, in 1975. She received the B.S. degree in electrical motors and automation from the Hefei University of Technology, Hefei, China, in 1996, and the Ph.D. degree in nuclear engineering from the Chinese Academy of Sciences, Hefei, in 2006.

She is currently a Professor with the International Thermonuclear Experimental Reactor Project, Institute of Plasma Physics, Chinese Academy of Sciences. Her current research interest includes power supply systems of fusion devices.

Zhengyi Huang was born in Anhui, China, in 1984. He received the B.S. degree in mechanical design manufacturing and automation from Anhui Polytechnic University, Wuhu, China, and the M.S. degree in electrical engineering from the Hefei University of Technology, Hefei, China, in 2018.

He is currently with the Institute of Plasma Physics, Chinese Academy of Sciences (ASIPP), Hefei. His current research interests include electromagnetic compatibility (EMC) analysis and dc busbar design.

Jie Zhang was born in Anhui, China, in 1988. He received the B.S. degree in mechanical design and automation from Anhui Agricultural University, Hefei, China.

He is currently an Engineer with the Institute of Plasma Physics, Chinese Academy of Sciences, Hefei. His research interests include mechanical design and finite-element analysis.

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